POWER PULSE

Symmetrical Components Important Formulae

Notes by Balaji Sir

1 Operator a

1. $$a = 1\angle 120^\circ = -\tfrac{1}{2} + j\tfrac{\sqrt{3}}{2}, \quad a^2 = 1\angle 240^\circ, \quad a^3 = 1, \quad 1 + a + a^2 = 0$$

2 Resolution

2. $$V_{a0} = \tfrac{1}{3}(V_a + V_b + V_c), \;\; V_{a1} = \tfrac{1}{3}(V_a + aV_b + a^2V_c), \;\; V_{a2} = \tfrac{1}{3}(V_a + a^2V_b + aV_c)$$
3. $$V_a = V_0 + V_1 + V_2, \quad V_b = V_0 + a^2V_1 + aV_2, \quad V_c = V_0 + aV_1 + a^2V_2$$
4. $$I_n = 3I_{a0}$$

neutral current

5. $$S = V_aI_a^* + V_bI_b^* + V_cI_c^* = 3\left(V_0I_0^* + V_1I_1^* + V_2I_2^*\right)$$

power is invariant with this scaling

6. $$\text{Voltage unbalance} = \frac{|V_2|}{|V_1|} \times 100$$

3 Sequence impedances

7. Lines: $Z_1 = Z_2, \quad Z_0 \approx 2\text{–}3.5\, Z_1$
8. Transformers: $Z_1 = Z_2 = Z_0$ (leakage), zero-sequence path depends on the connection
9. Neutral impedance $Z_n \Rightarrow 3Z_n$ in the zero-sequence network
10. Y–Δ: $+30^\circ$ (positive), -30° (negative), no zero-sequence through Δ

ANSI/IEEE convention

11. Positive: only source; negative and zero: no source